Study Questions: Systems

1. Pick up a flat crocheted coaster or dishcloth. Lay it on the table. It lies flat — zero Gaussian curvature. Now imagine adding two extra increases to every round. What would happen to the surface, and what kind of curvature would emerge? Why?

2. Daina Taimina's breakthrough was realizing that "too many increases" is not a mistake but a model of hyperbolic space. In your own crochet experience, have you ever produced a ruffled or wavy fabric unintentionally? What was happening topologically?

3. Consider a crocheted amigurumi sphere (a stuffed ball). The crocheter increases for the first half and decreases for the second half. Describe the Gaussian curvature at different points on the sphere. How does the Euler characteristic of this closed surface relate to the Gauss-Bonnet theorem?

4. Look at a small swatch of single crochet and think of it as a graph. Identify three stitches (nodes) and trace the edges (connections) between them. What is the **degree** of a stitch in the middle of the fabric versus a stitch at the edge?

5. When you crochet a shell stitch (5 dc in one stitch), you create a fan-out cluster in the stitch graph. How does this change the local connectivity compared to a row of plain single crochet? What is the degree of the base stitch that receives the shell?

6. Compare working in rows (back and forth) to working in the round. In graph theory terms, what is the key structural difference? How does this relate to the distinction between feedforward and recurrent neural networks?

7. A spike stitch reaches down past the previous row to insert into a stitch two or more rows below. In neural network terms, what is this analogous to? Why might a crocheter (or a network architect) want this kind of connection?

8. The standard flat circle formula for single crochet is 6 increases per round. Explain why this specific number produces zero curvature. What would happen with 5 increases per round? With 7?

9. When you hold your crochet work-in-progress, identify the **Markov blanket** as a topological boundary. Is this boundary an open curve or a closed curve? How does the answer change depending on whether you are working in rows or in the round?

10. Consider the crocheted fabric as a neural network. The foundation chain is the input layer, each subsequent row is a hidden layer, and the final row is the output layer. What "information" flows through this network, and how does each layer transform it?

11. A decrease stitch (sc2tog) merges two nodes from the previous layer into one node in the current layer. In neural network terms, this reduces dimensionality. When would a crocheter use this operation, and what is the topological effect on the fabric surface?

12. The Crochet Coral Reef project uses hyperbolic crochet to model reef organisms. Why is crochet particularly well-suited to modeling biological forms with negative curvature? What other construction methods fail at this task, and why?

13. Imagine crocheting a Mobius strip: you work a long chain, give it a half-twist, and join to crochet along both "sides" of the chain (which are now one continuous side). What topological property does this object have that a standard crocheted band does not? How many boundaries does it have?

14. In Active Inference, prediction error arises when observations do not match the generative model. Describe a specific crochet situation where the topology of the fabric generates prediction error for the crocheter. How might the crocheter resolve it?

15. Consider the Euler characteristic formula V - E + F. For a small crocheted flat disc (say, 3 rounds of single crochet), estimate V (stitches/nodes), E (connections/edges), and F (enclosed spaces/faces). Does the calculation yield the expected Euler characteristic for a disc?

16. A crocheted Klein bottle has no inside or outside — it is a non-orientable surface. If you were given instructions to crochet one, at what point in the construction would the non-orientability emerge? What stitch operation creates this topological twist?

17. How does the **clustering coefficient** of a crochet stitch graph change when you switch from a plain single crochet fabric to a shell-and-lace pattern? What visual and structural characteristics of the fabric correspond to high versus low clustering?
18. In a recurrent neural network, information from the output feeds back to influence future processing. When crocheting in the round without joining (spiral crochet), each round flows directly into the next with no clear boundary. How does this continuous spiral structure relate to the concept of recurrence? What are the implications for the Markov blanket?
19. The Lorenz manifold crocheted by Osinga and Krauskopf required precisely controlled increases and decreases to match the mathematical surface. In Active Inference terms, what served as the generative model, and how was prediction error measured and minimized during the construction?
20. Reflect on the entire module. You have learned that crochet fabric has topology, that the stitch network is a graph, and that neural networks share this graph structure. In what ways does this triple connection — topology, networks, and Active Inference — change how you think about the act of crocheting? Pick up a piece of your own crochet work and describe what you now see differently.